Describe a wave

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify amplitude, wavelength, frequency and period.

SUCCESS CRITERIA - I can...

1. Measure amplitude from equilibrium.
2. Measure wavelength between matching points.
3. Connect frequency and period.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “amplitude” and “wavelength” to explain your answer. Sentence starters are optional.

amplitude: maximum displacement from equilibrium for the wave

wavelength: distance between corresponding points on repeated waves

frequency: number of cycles each second

01 Warm-up: bring back an earlier idea

A sample warms by **3°C** after a certain energy input. If absorbed energy becomes **5** times as large with the same mass and material and no phase change, find the new rise.

Word help: absorbed = taken into the specified object or system

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

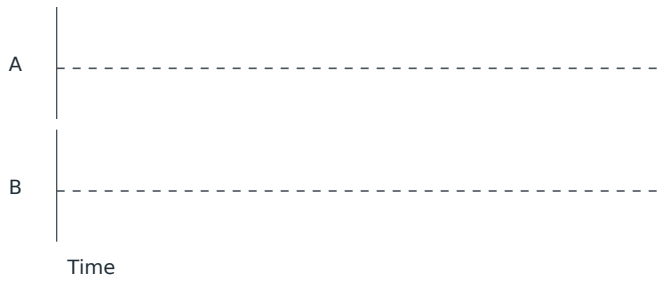
02 Worked example: explain the prediction

Reference: f = cycles/Δt; T = **1**/f.
f is frequency (Hz); T is period (s). λ is crest-to-crest wavelength; amplitude is maximum displacement.

Worked reasoning: A wave makes **6 cycles** in **3 s**: f=**2 Hz** and period T=**0.5 s**. Amplitude measures maximum displacement, not the spacing of crests.

Your turn: Draw one spatial wave and **label amplitude** and wavelength **0**. Separately mark one period on a displacement-time graph.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

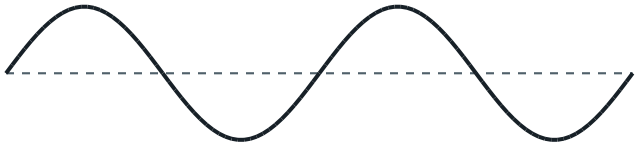


Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Label amplitude** and one wavelength on the wave diagram.
2. **Explain why crest-to-trough height** is not the amplitude.

Sentence starter: I think ... . I can point to ..., which shows ..., so ... .



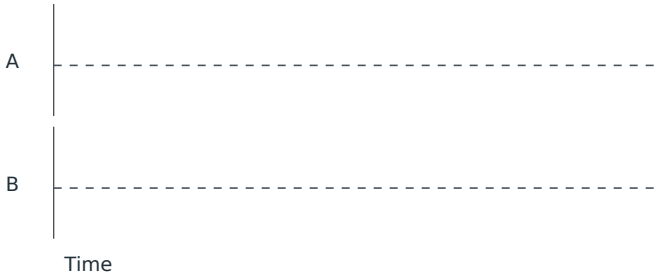
Sketch and label your response.

Marzano check - circle: ① ② ③

04 Apply the physics

A point on a rope completes **6 cycles** in **5 s**. **Find its frequency** and period. In a separate trial, each cycle takes **0.5 s**. Find that trial’s frequency. **Use two time** traces to **show which trial oscillates more** often.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

Describe a wave

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Does increasing amplitude necessarily increase frequency? Use the definitions to explain **0**.

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **4 kg** sample has c=**600 J**/(kg·°C) and warms by **8°C** without changing phase. **Find Q**.
Label temperatures, samples and relevant energy transfers.

Sentence starter: I found … [unit]. This value means … in this situation.

Sample A

Sample B

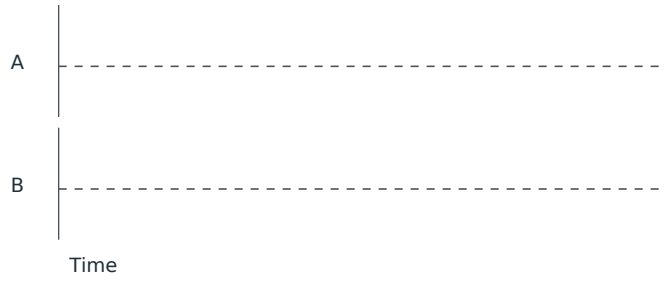
Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

07 Exit check

A wave has period **2 s**. **How many** cycles occur in **6 s**? **Find frequency**.

Sentence starter: I found … [unit]. This value means … in this situation.

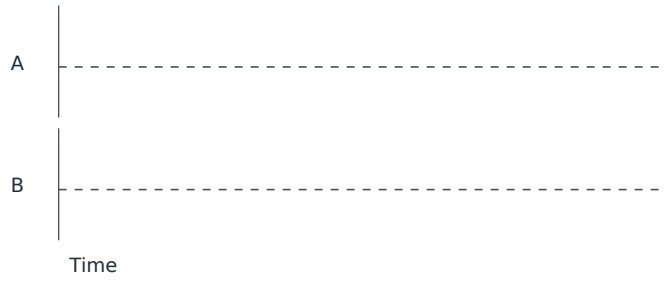


Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

1. **Draw two waves** with equal wavelength but different amplitudes.
2. **Explain which feature changes** and which does not.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: 1 2 3